

WILSON Fire Project — Validation & Benchmark Data

A. Hindcast validation against FIRED satellite perimeters

Thirteen end-to-end hindcast runs (NASA FIRED daily perimeters as ground truth, Open-Meteo ERA5 weather, Copernicus COP30 terrain + CORINE fuels). Differential Evolution calibrates five hidden parameters (moisture offset, wind multiplier, wind direction offset, ignition threshold, burn-time multiplier); budget `maxiter`=2, `popsiz`e=4; coarse grid 200×200 (~278 m), final IoU scored on 800×800 (~69 m). *Point* = seeded from day-1 polygon centroids (from-ignition skill). *Assim.* d_n = seeded from the observed cumulative perimeter of day n (FARSITE-style validation), predicting the next 24 h (48 h where noted).

Table 1: Hindcast IoU for all scenarios. A_p/A_t = predicted/truth area ratio; k_w = calibrated wind multiplier; θ_i = ignition threshold; b_t = burn-time multiplier.

Fire	Scenario	IoU (%)	A_p/A_t	k_w	θ_i	b_t	Interpretation
Rhodes 2023	Point, d_1	42.6	1.9	0.62	0.55	2.57	d_1 already 968 ha, spreading
Rhodes 2023	Assim. d_2	20.8	4.6	1.04	0.96	1.75	real $d_2 \rightarrow d_3$ growth tiny (+17%)
Rhodes 2023	Assim. d_4	47.6	1.8	0.74	0.80	1.78	main run of fire
Evoia 2021	Point, d_1	1.9	26	0.76	0.61	1.03	d_1 smolder (43 ha), unmatchable
Evoia 2021	Assim. d_2	9.4	9.6	2.20	0.16	1.66	$d_2 \rightarrow d_3$ eruption (43×)
Evoia 2021	Assim. d_4	53.2	1.1	1.41	0.91	3.46	eruption phase, well captured
Evoia 2021	Assim. d_4 , 48 h	52.9	1.6	2.53	0.72	4.14	2-day forecast holds up
Evoia 2021	Assim. d_6	66.7	1.4	1.13	0.88	1.03	steadier growth phase
Evoia 2021	Assim. d_8	90.2	1.1	1.10	0.76	1.46	late stage, small marginal growth
Evros 2023	Point, d_1	2.0	24	2.49	0.59	1.18	d_1 smolder (22 ha)
Evros 2023	Assim. d_4	43.0	2.2	1.59	0.47	4.55	explosive $d_4 \rightarrow d_5$ (3×)
Evros 2023	Assim. d_8	71.3	1.4	1.04	0.96	1.75	mid-late stage
Mean, assimilated d_4 (3 fires)		47.9					
Mean, all assimilated (9 runs)		50.6					
Mean, point d_1 (3 fires)		15.5					

Reading the sweep. Skill tracks the *relative* next-day growth of the real fire: seeding during eruption days (Evoia $d_2 \rightarrow d_3$: 43× area jump) or stall days (Rhodes $d_2 \rightarrow d_3$: +17%, while the unsuppressed model keeps running) hurts; mid/late-stage seeding reaches 43–90%. The model *always* over-predicts ($A_p/A_t > 1$ in every run): no suppression or evacuation effects are modelled, and a low DE budget was used. A repeat of Evoia d_4 at half the DE budget scored 51.6% vs. 53.2%, so results are not budget-limited. Note the cumulative-scoring convention includes the assimilated seed area, which inflates late-day IoU.

B. Historical fire context (FIRED + ERA5)

C. Fire-model physics characterization

Controlled synthetic landscapes (uniform fuel, flat unless noted, 50 m cells, $dt = 0.5$ min, 60-min fires, fixed RNG seed).

At the interactive default (800², $dt = 0.05$ min) the engine sustains ~63 steps/s ≈ 3.2 simulated minutes per wall second; the 200² optimizer grid runs ~1200 steps/s. WebSocket frame encoding (RLE GeoJSON, 80k fire cells on 800²) costs 7.3ms and 104KB per frame. Microclimate memory serializes to 1.4MB JSON in 25ms with a lossless round-trip.

Table 2: FIRED cumulative burned area (ha) by day, and ERA5 daily-mean weather at the ignition-day / day-4 seeds. Low ERA5 wind on eruption days (0.4–3 m/s) is why the optimizer’s wind multiplier and the $6\times$ upper bound exist: the 0.25° reanalysis misses mesoscale and pyroconvective winds.

Fire	Cumulative area (ha)				ERA5 (day 1 / day 4)		
	d_1	d_2	d_4	final	wind (m/s)	T ($^\circ\text{C}$)	RH (%)
Evoia 2021 (ev. 2871, 12 d)	43	130	11 226	52 521	1.5 / 0.4	27.9 / 34.0	61 / 22
Rhodes 2023 (ev. 130111, 10 d)	968	1 763	4 174	19 874	1.7 / 3.0	26.9 / 24.9	52 / 76
Evros 2023 (ev. 105620, 18 d)	22	1 789	17 928	89 422	1.5 / 2.5	22.7 / 24.3	85 / 55

Table 3: Fuel composition of each $0.5^\circ\times 0.5^\circ$ simulation domain (CORINE + OSM overlay, 200×200 grid; non-combustible includes sea for these coastal domains).

Fire	Non-comb.	Roads/urban	Pine	Maquis	Grass/phrygana	Oak
Evoia 2021	45.9%	16.5%	11.0%	6.7%	4.6%	2.4%
Rhodes 2023	58.5%	12.0%	6.4%	9.0%	8.6%	–
Evros 2023	47.7%	13.5%	6.1%	6.4%	10.9%	9.8%

Table 4: Rate of spread (dry grass, moisture 0.06). Left: vs. open wind (flat). Centre: vs. slope (no wind). Right: vs. moisture (6 m/s wind). Downwind ROS saturates above ~ 12 m/s open wind (Rothermel 1 500 ft/min effective-wind cap); upwind spread stays at the zero-wind 0.8 m/min backing rate. Moisture shows the η_m extinction cliff between 0.10 and 0.15 (grass m_x).

Wind (m/s)	ROS (m/min)	Slope ($^\circ$)	ROS $_{\uparrow}$ (m/min)	Moisture	ROS (m/min)
0	0.8	0	0.8	0.03	39.2
2	8.3	5	0.8	0.06	31.7
4	18.3	10	3.3	0.10	28.3
6	31.7	15	5.0	0.15	0.8
9	64.2	20	7.5	0.20	0.8
12	82.5	30	15.0	0.24	0.8
15	82.5				

Table 5: Fuel-dependent behaviour at 6 and 12 m/s open wind. The wind adjustment factor (WAF) reduces open wind to midflame wind under canopy; surface spread under pine/oak canopy is therefore minimal at ERA5-scale winds — crown-fire behaviour is absorbed by the optimizer’s wind and per-fuel multipliers instead.

Fuel	WAF	Midflame @6 m/s	ROS @6 (m/min)	ROS @12 (m/min)
Dry grass	0.40	2.40 m/s	31.7	82.5
Maquis dense shrub	0.20	1.20 m/s	18.3	45.0
Phrygana low scrub	–	–	17.5	–
Olive grove	–	–	6.7	–
Aleppo pine	0.10	0.60 m/s	0.8	1.7
Oak forest	0.12	0.72 m/s	0.8	3.3

Table 6: Numerical properties. Resolution test: identical 10×10 km domain, 2-h fire, 6 m/s wind, CFL-scaled $dt \propto \Delta x$. Areas agree within 3% between 100- and 50-m cells; the 25-m grid over-shoots ~40% because firebrand spotting distances are resolution-independent in metres but land more, finer ignition targets. Determinism: two runs with the same NumPy seed produce bit-identical states.

Grid	Cell (m)	dt (min)	Burned (ha)	Performance	ms/step
100 ²	100	1.00	677	200 ² grid	0.9
200 ²	50	0.50	696	400 ² grid	3.5
400 ²	25	0.25	992	800 ² grid	15.8

D. Intervention efficacy

Identical wind-driven grass fires (6 m/s, 300² grid); barrier placed 3 km downwind of ignition; burned area after 3 h vs. control.

Table 7: Burned area after 3 h. The ember-transport experiment (bottom) isolates why containment lines underperform: with spotting disabled the 0.95-strength line holds completely, so *all* line breaching is firebrand transport over the corridor — consistent with real ember-driven Mediterranean fires. Firebreaks block embers whose flight path crosses them; water drops on the head knock the fire back directly.

Scenario	Burned (ha)	Reduction
Control (no intervention)	1 581	—
Firebreak, 150 m corridor	750	53%
Firebreak, 250 m corridor	734	54%
Containment line, $s = 0.7$	1 493	6%
Containment line, $s = 0.95$	1 404	11%
Containment $s = 0.7$ + water + humidity	1 470	7%
3 aerial drops on fire head at 60 min	861	46%
Containment $s = 0.95$, spotting disabled	743	53% (line holds; fire never crosses)

E. Sandbox PSO optimizer

Particle-swarm calibration (12 particles × 8 iterations, 12-parameter space, synthetic 200² terrain, 2-h fire vs. a drawn truth polygon; 6 worker threads).

F. Software verification & project statistics

- Regression suite (optimizer state-freeze, canopy-factor persistence, cross-resolution fire restore, parameter application): **5/5 pass**.
- All Python modules byte-compile; WebSocket protocol handlers import cleanly.
- Determinism: bit-identical simulation states under a fixed RNG seed (Table in §C).
- Codebase: ~9 100 lines of Python (core model, pipelines, server) + 5 300-line web client; 23 Greek fuel classes (4 non-burning); 41 generated UI icons.
- Data assets: FIRED Greece 2000–2024 daily perimeters (22 MB GeoPackage, 24 867 features); 34 cached DEM/CORINE rasters.

Table 8: Best-so-far IoU per PSO iteration. The microclimate warm start (ignition-frequency prior from 50 previous optimizer simulations) accelerates early convergence but converges to a worse optimum here: the learned spatial prior biases the search toward previously-burned corridors. Useful for fast triage; disable for final calibration.

Run	Iteration								Final	Wall (s)
	1	2	3	4	5	6	7	8	IoU	
Fresh	.00	.37	.40	.40	.44	.45	.50	.52	0.556	149
Microclimate warm start	.00	.42	.46	.47	.47	.47	.47	.47	0.476	129

G. Defects found and fixed during this test campaign

1. Firebrand spotting ignored containment-line strength at the landing cell; brands now damped by local crew strength (embers beyond the corridor unaffected — Table §D reflects the fix).
2. Hindcast: ERA5 weather was fetched for the FIRMS query start date rather than the seed day whenever `ignition_day > 1`.
3. Hindcast: the perimeter-assimilation + tunable-persistence upgrade existed only in an unpushed working copy; re-implemented and validated (Evoia d_4 : 53.2% vs. 52.0% previously).
4. Cosmetic: results report printed physically impossible negative “effective midflame moisture” (clip was not applied in the printout).

H. Overall assessment

From-ignition (day-1) forecasting is only skilful when the fire is already large at first detection (Rhodes: 42.6%); smoldering-start fires are structurally over-predicted ($\sim 2\%$). Operational-style assimilated forecasting averages $\sim 48\%$ IoU at day 4 and 71–90% late-event, inside the 30–60% band typical of published FARSITE/FlamMap validations, with a consistent over-prediction bias attributable to missing suppression modelling and smooth ERA5 winds. The physics engine reproduces the qualitative Rothermel responses (wind power-law with saturation, slope amplification, moisture extinction cliff, canopy wind sheltering), is deterministic under a fixed seed, resolution-consistent to within 3% down to 50-m cells, and fast enough for interactive use.